NSSA322 Lab 1 - Enterprise Directory Services & Cross-Platform Trust

Activity 0: Review & Configure DNS, DHCP, and pfSense Relay

DNS configuration

- Forward lookup zones exist and are AD- and FreeIPA-integrated

```
[root@idm ~]# ipa dnsforwardzone-show windows.lab
  Zone name: windows.lab.
  Active zone: True
  Zone forwarders: 192.168.10.2
  Forward policy: first
[root@idm ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search linux.lab
nameserver 127.0.0.1
[root@idm ~]# cat /etc/resolv.conf | grep -E "search|ndots|options"
search linux.lab
[root@idm ~]#
```

- Reverse lookup zone(s) created or confirmed

```
Allow transfer: none;

Zone name: linux.lab.
Active zone: True
Authoritative nameserver: idm.linux.lab.
Administrator e-mail address: hostmaster.linux.lab.
SOA serial: 1789690381
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant LINUX.LAB krb5-self * A; grant LINUX.LAB krb5-self * AAAA; grant LINUX.LAB krb5-self *
                    SSHFP;
Dynamic update: True
Allow query: any;
Allow transfer: none;
----------------------------
umber of entries returned 2
----------------------------
student@idm ~]$ dig @192.168.20.6 linux.lab

 <<>> DiG 9.16.23-RH <<>> @192.168.20.6 linux.lab
 (1 server found)
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 11425
; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

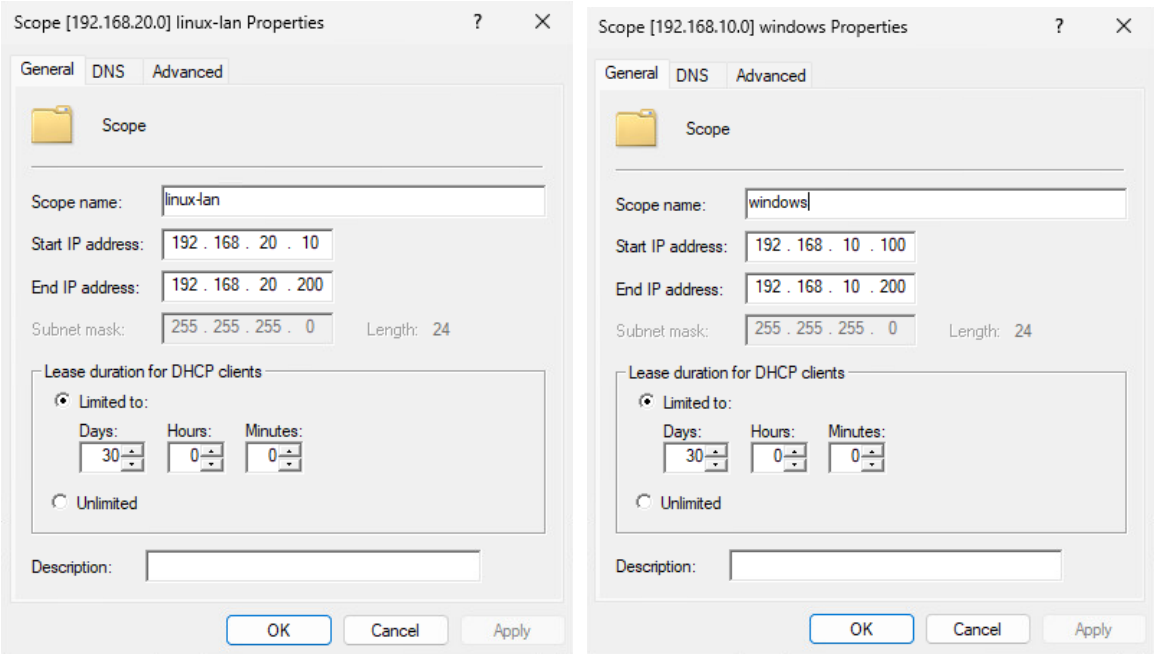
; OPT PSEUDOSECTION:
 EDNS: version: 0, flags:; udp: 1232
 COOKIE: 4d54a6f3a6714a6f010000006aac86a36a183e7522b65374 (good)
; QUESTION SECTION:
inux.lab.                      IN      A

; AUTHORITY SECTION:
inux.lab.               3600    IN      SOA     idm.linux.lab. hostmaster.linux.lab. 1789690381 3600 900 1209600 36
0

; Query time: 5 msec
; SERVER: 192.168.20.6#53(192.168.20.6)
; WHEN: Thu Sep 17 20:32:35 EDT 2026
; MSG SIZE  rcvd: 117

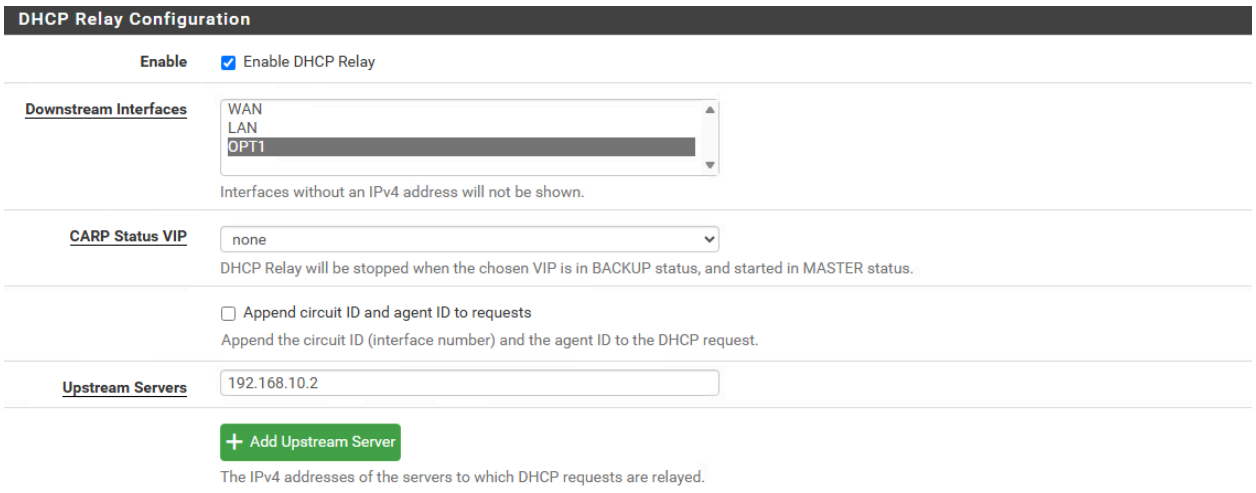
student@idm ~]$
```

DHCP configuration



- Scope(s) created to cover the RLES lab environment

pfSense DHCP relay



- DHCP Relay Configuration screen.

Verification: Windows client

```
Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : windows.lab
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-50-56-B0-EE-F6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::f619:463c:2e35:6fe8%3(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.102(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, October 15, 2026 1:20:25 PM
   Lease Expires . . . . . . . . . . : Saturday, November 14, 2026 1:20:24 PM
   Default Gateway . . . . . . . . . : 192.168.10.254
   DHCP Server . . . . . . . . . . . : 192.168.10.2
   DHCPv6 IAID . . . . . . . . . . . : 100683862
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-62-BF-EF-00-50-56-B0-EE-F6
   DNS Servers . . . . . . . . . . . : 192.168.10.2
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

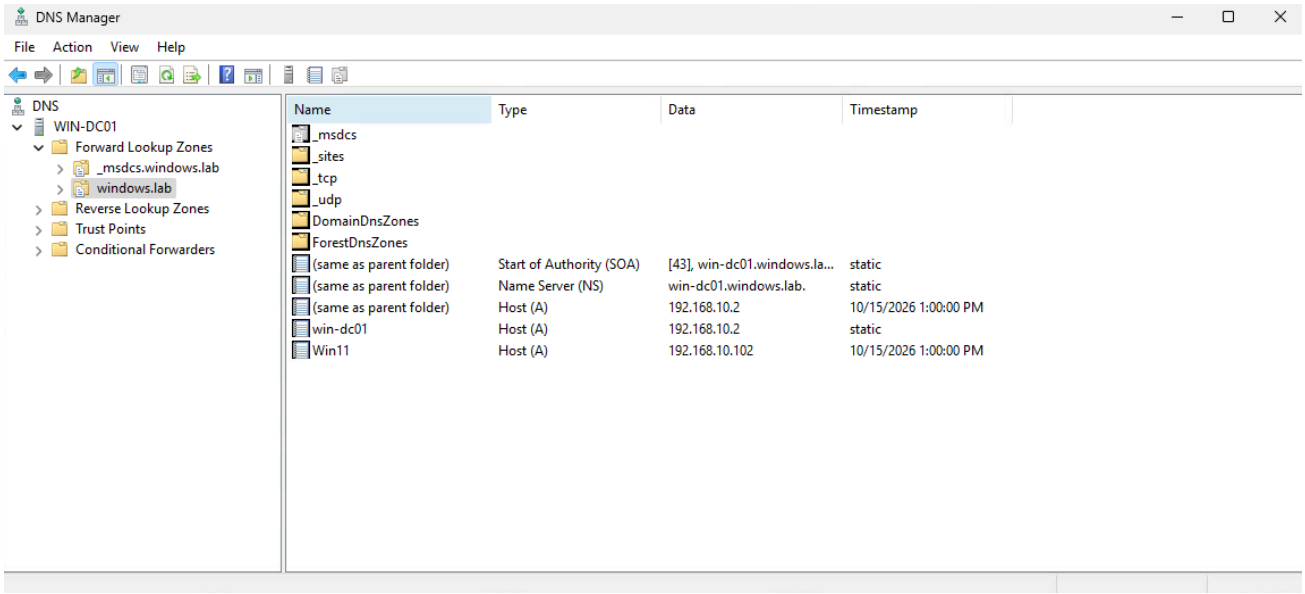
- Windows client ipconfig /all result

Verification: Rocky Linux client (RLES)

```
[root@rocky1 ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:50:56:b0:5b:3e brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    inet 192.168.20.13/24 brd 192.168.20.255 scope global dynamic noprefixroute ens192
       valid_lft 2574153sec preferred_lft 2574153sec
    inet6 fe80::250:56ff:feb0:5b3e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@rocky1 ~]#
```

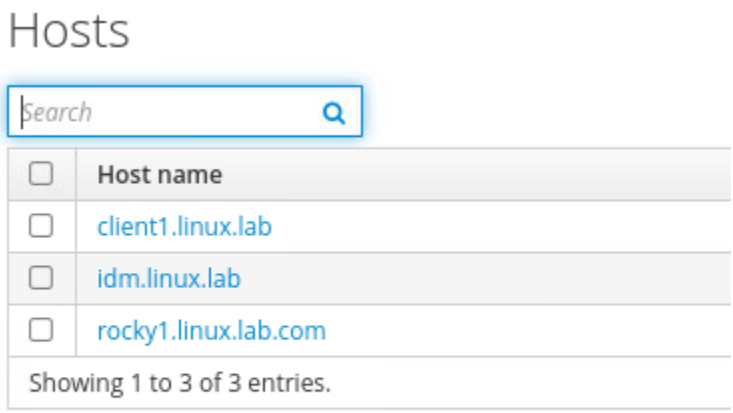
- Rocky Linux client ip configuration.

Verification: DNS records



- Client records appear in Windows DNS

Verification: IDM records



- Client records appear in Linux IDM.